The registered SPECTRUM arm: all 28 entering model × wording-frame × prompt-order conditions

A Does the companion's compliance predict B's loss?
fitted slope β, simultaneous 95% interval

19 of 28 positive · 4 negative · 5 span zero

B Does A sit above, below or on that relation?
δ_difficulty, joint 95% decision interval

17 within ±10 pp (L3e) · 1 above prediction (L3d) · 10 large and negative, no level

Qwen3-4B
B first: f0, f1, f2, f3
B second: f0, f1, f2, f3

Qwen3-4B-Instruct-2507
B first: f1, f3
B second: f1, f3

Gemma-3n-E2B-it
B first: f0, f1, f2, f3
B second: f0, f1, f2, f3

Gemma-3n-E4B-it
B first: f0, f1, f2, f3
B second: f0, f1, f2, f3

0.0 0.1 0.2 0.3

β (pp of B-loss per unit of companion compliance ι)
β > 0 means B fares WORSE beside an EASIER companion

−30 −20 −10 0 10

δ_difficulty = observed D(B←A) − loss predicted at ι_A, pp
negative: A costs B LESS than its compliance predicts

- B stated first
- B stated second
- β > 0: B worse beside an easier companion (the majority)
- β < 0: the intuitive direction — harder companions cost more
- interval spans zero
- L3e: A within ±10 pp of the prediction
- L3d: A above the prediction, interval excludes zero
- no registered level: residual large and negative
- ±10 pp registered equivalence region